Class 36: Magnetic Properties

Magnetic properties are commonly used in a variety of technologies. Audio speakers, and motors are commonplace examples of technologies that use magnets. More exotic technologies that use magnetism are the Magnetic Resonance Imaging (MRI) scanners, used in the medical field.

Magnetism is a phenomenon that has been known and used long before any understanding was developed on the science behind the phenomenon. Magnetism is observed in the form of an attractive force that some materials exert on some others, that is in addition to any other forms of attraction such as electrostatic attraction. While originally it was discovered as existing in some materials as obtained in nature, it was later discovered that moving electrical charges create a magnetic field. This discovery led to the manufacture of electromagnets where coils of current carrying conductors generate magnetic fields that are then used to serve specific purposes. Industrially, the use of electricity to generate magnetism, is the most commonly observed usage of magnetism, since it enables considerable control on the phenomenon.

The common equations associated with magnetism are listed below:

The externally generated magnetic field, using a coil carrying electricity is indicated by:

$$H = \frac{NI}{L}$$

Where, N is the number of turns of the coil, I is the current flowing through it, and L is the length of the coil.

The externally generated magnetic field can induce a magnetic field in another material, which is placed within the coil, and this is given by:

$$B = \mu H$$

Where μ is the permeability of the material.

In case the coil is in vacuum, the field in the vacuum is given by:

$$B = \mu_0 H$$

Where μ_0, is the permeability of vacuum.

The ratio

$$\frac{\mu}{\mu_0} = \mu_r$$

Is called relative permeability and is therefore without units.

The response of a sample to the externally generated magnetic can be thought of as a response that is in addition to the response of vacuum to the same field. This additional response, which is the extent of internal reinforcement or opposition to the applied field, is referred to as magnetization, denoted by M. Therefore the response of the material can be written as:

$$B = \mu H = \mu_0 H + \mu_0 M$$

And since the magnetization $M$is itself generated in response the external field, it can be related to the external field through:

$$M = \chi H$$

Where χ is called the susceptibility of the material.

Therefore

$$B = \mu_0 H + \mu_0 M = \mu_0 H + \mu_0 \chi H$$

Therefore:

$$\mu H = \mu_0 H(1 + \chi)$$

And

$$\frac{\mu}{\mu_0} = \mu_r = 1 + \chi$$

Or

$$\chi = \mu_r - 1$$

Since μ_r can be less than or greater than 1 (since some materials will oppose applied magnetic fields some others may augment the applied magnetic field), χ can be positive or negative.

Figure 36.1 below shows a schematic of the response of specific types of materials to externally applied magnetic fields. Diamagnetic materials weakly oppose the applied field ($\mu_r < 1, \chi$ is a small negative number), paramagnetic materials weakly augment the applied field ($\mu_r > 1, \chi$ is a small positive number), while ferromagnetic materials strongly augment the applied field ($\mu_r \gg 1, \chi$ is a large positive number)

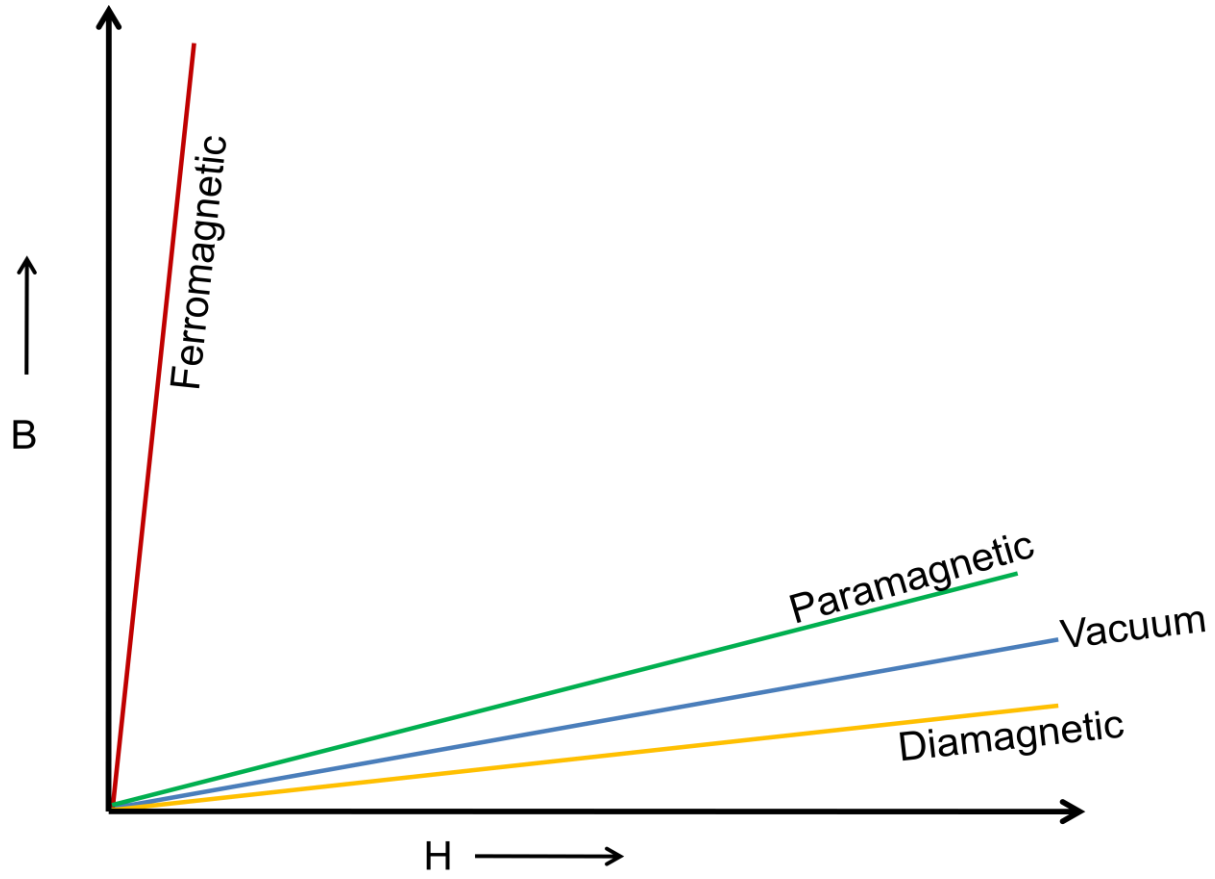


Figure 36.1: The response of specific materials to an externally applied magnetic field

In the theories for materials we have developed so far, we noted that the density of occupied states, as a function of energy, has a profile as shown in Figure 36.2 below.

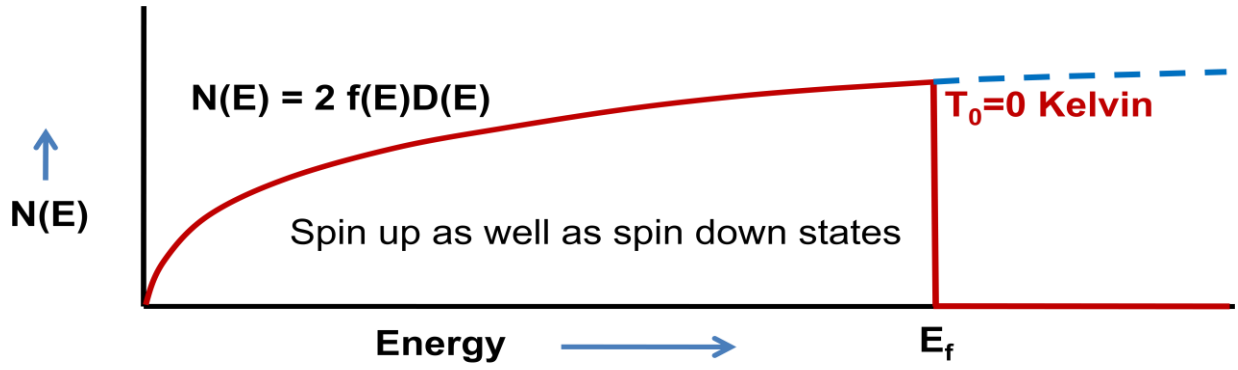


Figure 36.2: Density of occupied states as a function of energy. Both spin up as well as spin down states are included in the figure together.

The occupied states consist of electrons that are spin up and spin down. The spin of the electrons contributes to the magnetic behavior of the material. The density of occupied states of the material can therefore be redrawn as shown in Figure 36.3 below, where the axis have been interchanged and the electrons with spin down appear on one side of the figure, while the electrons with spin up, appear on the other side of the figure, as shown below.

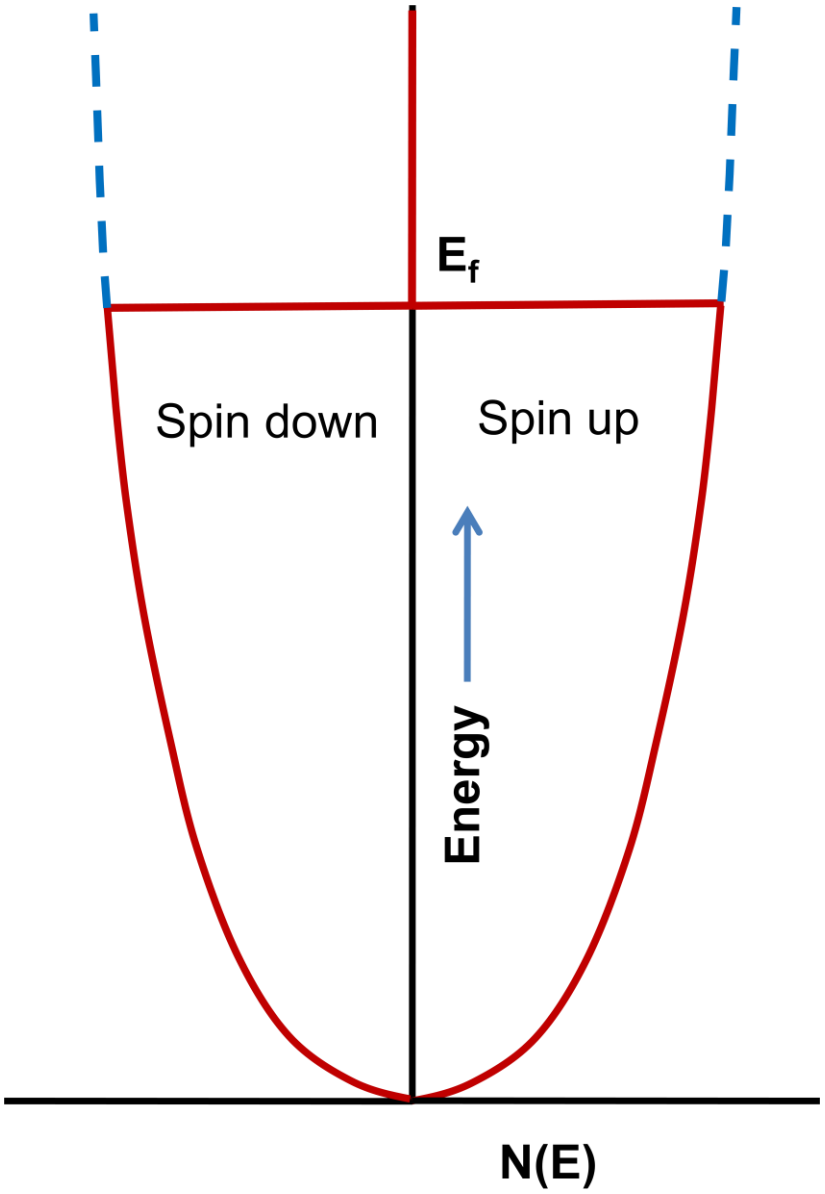


Figure 36.3: Density of occupied states as a function of energy, with the electrons having spin up and electrons having spin down, being shown separately. For the sake of clarity, the axis have been interchanged with respect to the earlier figure (36.2), and therefore the figure is rotated by 90°, counter clockwise, with respect to the earlier figure (36.2).

As an aside, it is important to note that although the form of Figure 36.3 above is similar to the $E\ Vs\ k$ diagram, they are not the same.

When a ferromagnetic material experiences an externally applied magnetic field, the states with spin aligned in the direction of the applied magnetic field, attain lower energies, while the states with spin opposed to the applied magnetic field, attain a higher energy state. Temporarily the situation can be thought of as shown in Figure 36.4(a) below. Since the Fermi energy of the system has to be a uniform value for the system, electrons move from the states with spin opposed to the magnetic field, to states with spin aligned with magnetic field, till the Fermi energy reaches a uniform value, as shown in Figure 36.4(b) below.

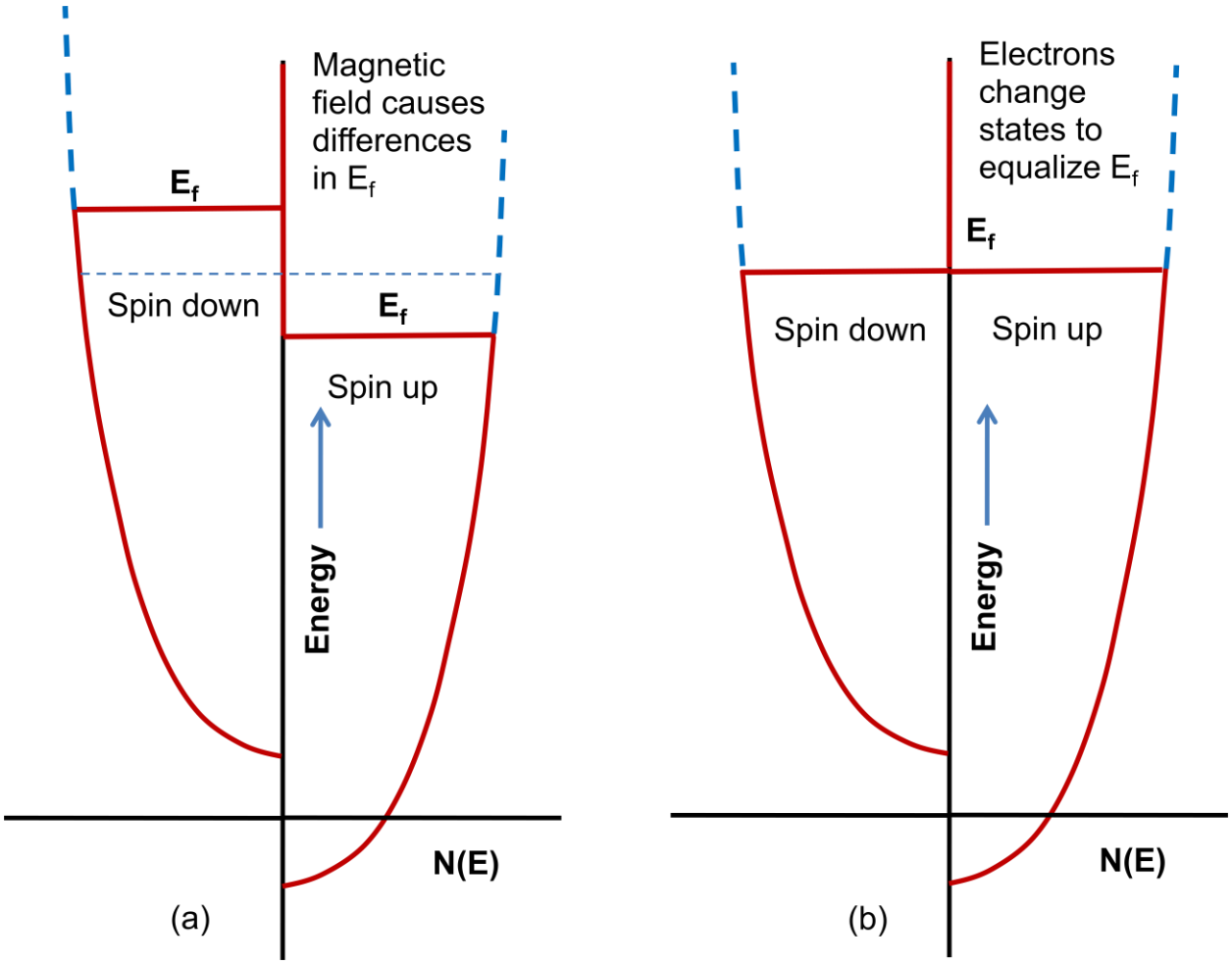


Figure 36.4: Response of a ferromagnetic material to an externally applied magnetic field: (a) Intermediate state where the states with spin aligned in the direction of the applied magnetic field, attain lower energies, while the states with spin opposed to the applied magnetic field, attain a higher energy; (b) The final state of the system after electrons change states to attain a uniform Fermi energy for the system.

This increase in the number of electrons with spin aligned favorably with the applied magnetic field, explains the Magnetization behavior of materials, M.

The theories we have developed thus far, are therefore able to explain magnetic behavior of materials.